

ML Graded Quiz M5

Question 1

Using the below data to train a Support Vector Classifier:

$$X = \begin{bmatrix} 1 & 8 \\ 4 & 4 \\ 1 & 2 \\ 8 & 4 \\ 11 & 1 \\ 12 & 8 \end{bmatrix}$$

with corresponding target class

$$y = \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

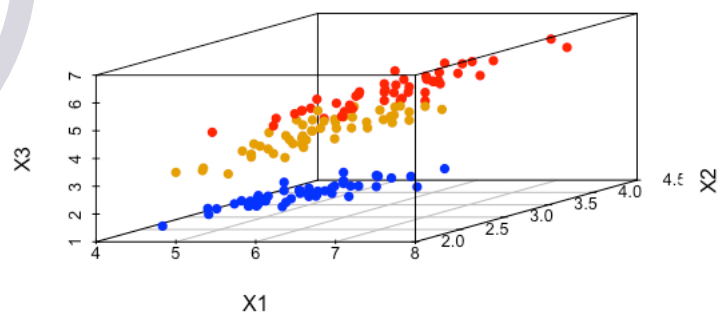
The predicted outcome for an observation [6, 6] would be:

- ☐ Class 1 using a polynomial Kernel with degree = 3
- ☐ Class -1 using a polynomial Kernel with degree = 2
- ☐ Class -1 using a polynomial Kernel with degree = 3
- ☒ Class 1 using a polynomial Kernel with degree = 1

Question 2

Which neural network architecture would you choose for the below data where X1, X2, and X3 are the predictors and target distribution shown?

(Assume a Softmax activation function and probabilities of all target classes are required from the model.)



- ☐ 3 input nodes and 1 output nodes
- ☐ 3 input nodes and 2 output nodes
- ☐ 3 input nodes and 3 output nodes
- ☒ 2 input nodes and 3 output nodes

Question 3

The difference between PCA and kernels is:

- ☐ Kernels and PCA increase feature dimensions in different ways.
- ☐ Kernels and PCA reduce feature dimensions in different ways.
- ☐ Kernels create additional feature dimensions whereas PCA reduces feature dimensions.
- ☒ PCA creates additional feature dimensions whereas kernels reduces feature dimensions.

Question 4

Which activation function suffers from the vanishing gradient problem?

- ☐ adam
- ☐ relu
- ☐ softsign
- ☒ tanh

Question 5

Suppose RBF, Sigmoid, Polynomial, and linear kernels produced the same performance metric. If maximum computational speed was the only other criteria for selecting a kernel, you would choose _____.

- ☐ a linear kernel
- ☐ the RBF Kernel
- ☐ the Polynomial kernel
- ☒ the Sigmoid kernel

Question 6

Which activation function should be used in the output layer for a classification problem of more than 2 classes?

- ☐ tanh
- ☐ softmax
- ☐ relu
- ☒ binary crossentropy

Question 7

Which of the below is **true**?

- ☐ The number of operations required to compute the inner product for the inputs of a kernel will decrease significantly with fewer observations.
- ☐ The number of operations required to compute the inner product for the inputs of a kernel will always increase with an increase in features.
- ☐ The number of operations required to compute the inner product for the inputs of a kernel is not necessarily proportional to the number of features.

- ☒ The number of operations required to compute the inner product for the inputs of a kernel will increase significantly with more observations.

Question 8

Which function sends feedback signals on how to adjust the weights?

- ☐ optimizer
- ☐ tanh function
- ☐ activation function
- ☒ loss function

Question 9

In a one-dimensional space, we have two points, namely $x = 2$ and $z = 4$. Assume a polynomial kernel of degree 2 and scaling parameter of 1 with zero offset. What is the kernel value?

- ☐ 8
- ☐ 81
- ☐ 64
- ☒ 3

Question 10

Which of the following refers to the architecture of the neural network?

- ☐ Number of observations
- ☐ Bias
- ☐ Number of hidden layers
- ☒ Weights

Question 11

The hyperparameter that controls the penalty for misclassifications is known as

- ☐ the margin.
- ☐ the regularization parameter.
- ☐ the kernel.
- ☒ the gamma parameter.

Question 12

Suppose we have a vector, $X = [0.45, 0.025, 0.15]$, of input into a neuron with associated weights $w = [0.33, 0.33, 0.34]$ and bias = 1. Suppose the activation function at that neuron is the sigmoid function. What output can we expect at that neuron?

- ☒ 0.2300

- ☐ 1.000
- ☐ 0.7699
- ☒ 1.299

Question 13

In a one-dimensional space, we have two points, namely $x = 2$ and $z = 4$. Assume a polynomial kernel of degree 2 and scaling parameter of 1 with offset = 1. What is the kernel value?

- ☐ 73
- ☐ 16
- ☐ 64
- ☒ 81

Question 14

How many hidden layers are in this neural network architecture?

```
tf.random.set_seed(36)

model_1 = tf.keras.Sequential([

    tf.keras.layers.Dense(8, activation='sigmoid', input_shape=(1,)),

    tf.keras.layers.Dense(4, activation='sigmoid'),

    tf.keras.layers.Dense(1)

])

model_1.compile( loss = tf.keras.losses.mae,

    optimizer = tf.keras.optimizers.Adam(),

    metrics = ['mae'])

model_1.fit( X_train, y_train, epochs = 100, batch_size = 32)
```

- ☐ 4
- ☐ 2
- ☐ 3
- ☒ 1

Question 15

Suppose you are working on a classification problem. You fitted an SVM classifier to your training data and found the accuracy for your training set to be 88%; however, the accuracy on your test set is 70%. Which of the following should you do?

- ☐ Decrease the penalty strength c .

1

QUESTION 15

- | | |
|----|----|
| 1 | 2 |
| 5 | 6 |
| 9 | 10 |
| 13 | 14 |

- ☐ Increase the gamma parameter to consider points farther away from the decision boundary.
- ☐ Increase the penalty strength c , since it will classify more unseen data correctly.
- ☒ Decrease the gamma parameter to consider points closer to the decision boundary.

Question 16

Which activation function suffers from the vanishing gradient problem?

- ☐ sigmoid
- ☐ adam
- ☐ relu
- ☒ softsign

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